

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1684

Gaston 345 kV
February 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1684. The Interconnection Customer proposes a 459.1 MW wind facility interconnecting at the Gaston 345 kV substation in the North Carolina area. The request was received on March 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 402 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,524,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Tower Tap / Cedar Falls Wi Substation 138.0 kV line	N Tower Tap / Haggin 345.0 kV Line	123.4	304	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Tower Tap / Cedar Falls Wi 138.0 kV	Rebuild line with higher-capacity conductor	\$7,973,000	19% (\$1,524,000)
	Total allocated to project		\$1,524,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.